HUM 4201: PRINCIPLES OF ECONOMICS

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Chapter-02: Measuring the Cost of Living

MOTIVATION

Babe Ruth vs Modern Players

- In **1931**, during the Great Depression:
  - **Babe Ruth,** a famous Baseball player, earned **$80,000**
  - President **Herbert Hoover** earned **$75,000**
- When asked about earning more than the president:
  - Ruth replied: *"I had a better year."*
- In **2015**:
  - Average major league Baseball (MLB) salary ≈ **$4 million**
  - **Clayton Kershaw** earned **$31 million**
- Initial impression:
  - Huge increase in salaries over time
- But:
  - **Prices of goods and services have also risen**

The Core Economic Problem

- In 1931:
  - ✓ A **nickel** could buy an ice-cream cone
  - ✓ A **quarter** could buy a movie ticket
- Today:
  - ✓ Prices are much higher
- ❓ Key Question:
  - ✓ Did Babe Ruth enjoy a higher or lower **standard of living** than modern players?

MOTIVATION

From Nominal Values to Real Meaning

- Comparing money across time requires adjusting for price changes
- **Problem:**
  - ✓ Dollar values alone (**nominal values**) are misleading
- **Solution:**
  - ✓ Convert into **purchasing power**
- Economists need a way to:
  - ✓ Compare **income across time**
  - ✓ Adjust for **changes in price levels**
- 👉 This leads to:
  - ✓ **Consumer Price Index (CPI)**

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The Consumer Price Index (CPI)

- A measure of the **overall cost of goods and services** bought by a typical consumer
- Computed by: **Bureau of Labor Statistics (BLS)**
- Frequency: **Monthly**

Purpose of CPI

- Tracks **changes in cost of living over time**
- Measures **inflation rate**
- Used for:
  - Economic policy
  - Wage adjustments
  - Indexation

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How CPI is Calculated – Overview

The BLS follows **5 key steps**:

1. Fix the basket
2. Find prices
3. Compute basket cost
4. Choose base year & compute index
5. Compute inflation rate

👉 We now examine each step in detail

Step 1 – Fix the Basket

- To measure the cost of living, **more weight** is given to prices of items the **typical consumer buys most.**
- If **consumers buy more hot dogs** than hamburgers, hot dog prices matter more.
- The BLS determines these weights by surveying consumers about their typical basket of goods; for example, **4 hot dogs and 2 hamburgers.**

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Step 2 – Find Prices

- Collect prices of goods in different years
- **Example:**

Year	Hot Dog Price	Hamburger Price
2016	$1	$2
2017	$2	$3
2018	$3	$4

👉 Thousands of prices collected in reality

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Step 3 – Compute Basket Cost

- Multiply prices by quantities in basket
- **Calculations:**

  2016

  = (4×\$1) + (2×\$2) = **\$8**

  2017

  = (4×\$2) + (2×\$3) = **\$14**

  2018

  = (4×\$3) + (2×\$4) = **\$20**
- Basket is fixed → only prices change
  - Holding basket constant:
    - ✓ Isolates **price changes**
    - ✓ Ignores **quantity changes**
  - This is crucial for measuring inflation

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Step 4 – Choose Base Year & Compute CPI

- Select a **base year** (benchmark)
- Example:
  - Base year = **2016**
  - Basket cost = **$8**
- **CPI Formula:**

$$CPI = \frac{Price\ of\ basket\ in\ current\ year}{Price\ in\ base\ year} \times 100$$

CPI Calculation Example

2016 (Base Year):

$$(8/8) \times 100 = 100$$

2017:

$$(14/8) \times 100 = 175$$

2018:

$$(20/8) \times 100 = 250$$

Interpretation:

- CPI 175 → prices are **75% higher than base year**

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Step 5 – Compute Inflation Rate

Inflation Rate:

- ✓ *The percentage change in the price index from the preceding period*

Formula:

$$\text{Inflation } Rate = \frac{CPI_t - CPI_{t-1}}{CPI_{t-1}} \times 100$$

Inflation Calculation Example

2017:

$$(175 - 100)/100 \times 100 = 75\%$$

2018:

$$(250 - 175)/175 \times 100 = 43\%$$

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Step 1: Survey Consumers to Determine a Fixed Basket of Goods

Basket = 4 hot dogs, 2 hamburgers

Step 2: Find the Price of Each Good in Each Year

Year	Price of Hot Dogs	Price of Hamburgers
2016	\$1	\$2
2017	2	3
2018	3	4

Step 3: Compute the Cost of the Basket of Goods in Each Year

2016	(\$1 per hot dog × 4 hot dogs) + (\$2 per hamburger × 2 hamburgers) = \$8 per basket
2017	(\$2 per hot dog × 4 hot dogs) + (\$3 per hamburger × 2 hamburgers) = \$14 per basket
2018	(\$3 per hot dog × 4 hot dogs) + (\$4 per hamburger × 2 hamburgers) = \$20 per basket

Step 4: Choose One Year as a Base Year (2016) and Compute the CPI in Each Year

2016	(\$8/\$8) × 100 = 100
2017	(\$14/\$8) × 100 = 175
2018	(\$20/\$8) × 100 = 250

Step 5: Use the CPI to Compute the Inflation Rate from Previous Year

2017	(175 – 100)/100 × 100 = 75%
2018	(250 – 175)/175 × 100 = 43%

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Construction of CPI – Bangladesh Context

Institutional Framework

- ❖ CPI compiled by: **Bangladesh Bureau of Statistics (BBS)**
- ❖ International standard: Endorsed by:
  - ✓ **United Nations Statistical Commission (2020)**
- ❖ CPI is **compiled on a monthly basis**
- ❖ Uses:
  - ✓ Monthly prices of various items
- ❖ Annual CPI:
  - ✓ Calculated as **average of 12 monthly indices**
- ❖ Reference groups:
  - ✓ **Urban households**
  - ✓ **Rural households**

Types of CPI in Bangladesh

- ➢ Three principal CPI indices:
  1. **National CPI**
  2. **Urban CPI**
  3. **Rural CPI**
- ➢ National CPI: Derived by combining:
  - Urban CPI
  - Rural CPI
    - ✓ Using **appropriate weights**

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Construction of CPI – Bangladesh Context

Classification of CPI Basket

- All goods and services are classified into **12 divisions**:
    1. Food & Non-alcoholic Beverages
    2. Alcoholic Beverages, Tobacco & Narcotics
    3. Clothing & Footwear
    4. Housing, Water, Electricity, Gas & Fuels
    5. Furnishings & Household Maintenance
    6. Health
    7. Transport
    8. Communication
    9. Recreation & Culture
    10. Education
    11. Restaurants & Hotels
    12. Miscellaneous Goods & Services

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Construction of CPI – Bangladesh Context

Selection of Items

- Based on:
  - ✓ **Household Income and Expenditure Survey (HIES) 2016–17**
- Base index:
  - ✓ **2021–22 = 100**
- Data collection:
  - ✓ From **selected shops, markets, and service providers**

Price Collection System

- Total markets:
  - **154 markets**
    - ✓ 64 rural
    - ✓ 90 urban
- Special focus:
  - Dhaka & Chattogram city corporations
- Each market:
  - **3 outlets**
- 👉 For each item:
  - **3 price quotations collected**

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Construction of CPI – Bangladesh Context

Coverage of Items

- Food items: **242**
- Non-food items: **507**
- Total items: **749 goods and services**
- Prices collected using: **Structured schedules**
- Base year: **2021–22 = 100**

Concept of Weights

- CPI is a **weighted average of prices**
- Weights represent share of each item in **total household consumption**
- Determines:
  - ✓ Impact of price change on CPI
- Example:
  - ✓ Food price ↑ → large effect (high weight)
  - ✓ Recreation price ↑ → small effect (low weight)
- **Source of Weights: HIES 2016–17**

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Construction of CPI – Bangladesh Context

Group	Rural Weight	Urban Weight		Group	Rural Weight	Urban Weight
Food & Non-alcoholic Beverages	48.65	38.37		Transport	7.33	12.88
Alcoholic Beverages, Tobacco and Narcotics	2.97	2.08		Communication	2.26	2.47
Clothing and Footwear	6.36	5.69		Recreation & Culture	1.56	1.51
Housing, Water, Electricity, Gas, and Other Fuels	13.02	19.02		Education	3.20	4.71
Furnishings & Household Maintenance	3.94	3.68		Restaurants & Hotels	2.23	2.06
Health	4.55	3.97		Miscellaneous Goods & Services	3.92	3.56

- ❖ **National CPI Construction:** Weighted combination of urban & rural CPI
- ❖ Weights based on: Population & consumption structure

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Problems in Measuring the Cost of Living

Objective of CPI

- The goal of the consumer price index is to **measure changes in the cost of living**.
- CPI tries to gauge **how much incomes must rise** to maintain a **constant standard of living**.
- However, CPI is **not a perfect measure** of the cost of living.

Major Problems with CPI

- Three widely acknowledged problems:
    1. **Substitution Bias**
    2. **Introduction of New Goods**
    3. **Unmeasured Quality Change**
- These problems are **difficult to solve**.

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Problems in Measuring the Cost of Living

1. Substitution Bias

- Prices do not change proportionately across goods.
- Some prices **rise more than others**.
- Consumers respond by:
  - ✓ Buying **less of relatively expensive goods**
  - ✓ Buying **more of relatively cheaper goods**
- This behavior is called **substitution**.

☑ CPI assumes a **fixed basket of goods**.
☑ It ignores consumer substitution behavior.
☑ As a result, CPI **overstates the increase in cost of living**.

Example

- Base year:
  - ✓ Apples cheaper → consumers buy more apples
- Next year:
  - ✓ Pears become cheaper → consumers buy more pears
- CPI issue:
  - ✓ Still assumes **same quantity of apples purchased**
- Result:
  - ✓ CPI shows **higher cost of living than actual experience**.

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Problems in Measuring the Cost of Living

2. Introduction of New Goods

- New goods increase **consumer choice and variety**.
- More variety makes each dollar **more valuable**.
- Therefore, cost of maintaining same well-being **decreases**.

Example

- Compare:
  - ✓ $100 gift card (large store, wide variety)
  - ✓ $100 gift card (small store, limited variety)
- Consumers prefer **greater variety**
- Conclusion: More choices → **higher utility per dollar**

New Goods (CPI Problem)

- CPI uses a **fixed basket.**
- It does **not reflect increased value from new goods.**
- Therefore, CPI **fails to capture reduction in cost of living**.

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Problems in Measuring the Cost of Living

New Goods (Example: iPhone)

- iPhone introduced in 2025:
  - ✓ More portable and versatile than previous devices
- Effects:
  - ✓ Increased consumer opportunities
  - ✓ Reduced cost of achieving same satisfaction
- CPI issue:
  - ✓ Did not initially reflect this improvement
- Later:
  - ✓ Basket updated, but **initial benefit never captured**

3. Unmeasured Quality Change

- Quality changes affect **value of money**:
  - Lower quality → value of dollar falls
  - Higher quality → value of dollar rises
- CPI attempts to adjust for quality changes.

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Problems in Measuring the Cost of Living

3. Unmeasured Quality Change

- **Example:**
  - ✓ Car with better horsepower or fuel efficiency
- BLS adjusts price:
  - ✓ To reflect **constant quality comparison**
- Goal:
  - ✓ Measure price of **same-quality basket over time**

- **Problem**
  - Quality is **difficult to measure accurately**
  - Despite adjustments:
    - ✓ Measurement errors persist
  - Result:
    - ✓ CPI may still be **biased**

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The GDP Deflator versus the Consumer Price Index

GDP Deflator

- ❑ The GDP deflator is the **ratio of nominal GDP to real GDP**.
- ❑ Nominal GDP: Current output valued at **current prices**
- ❑ Real GDP: Current output valued at **base-year prices**
- ❑ Therefore: GDP deflator reflects the **current level of prices relative to the base year**.

Why Compare GDP Deflator and CPI?

- Economists and policymakers monitor:
  - ✓ **GDP deflator**
  - ✓ **Consumer Price Index (CPI)**
- Purpose: To gauge **how quickly prices are rising (inflation)**
- Usually: Both statistics **tell a similar story**

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The GDP Deflator versus the Consumer Price Index

Key Differences Overview

➢ Two important differences:
   1. **Coverage of goods and services**
   2. **Weighting method (fixed vs changing basket)**

Difference 1 – Coverage

- GDP deflator:
  - ✓ Includes prices of **all goods and services produced domestically**
- CPI:
  - ✓ Includes prices of **all goods and services bought by consumers**

Example 1 – Domestic Production

- Airplane produced by Boeing and sold to Air Force:
  - ✓ Included in **GDP**
  - ✓ Not included in **CPI basket**
- Result:
  - ✓ Price increase affects:
    - **GDP deflator ✓**
    - **CPI ✗**

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The GDP Deflator versus the Consumer Price Index

Difference 1 – Coverage

Example 2 – Imported Goods

- Volvo cars (produced in Sweden):
  - Not part of U.S. GDP
  - Part of **consumer purchases**
- Result:
  - Price increase affects:
    - **CPI ✓**
    - **GDP deflator ✗**

Soybean Price Example (BD Context)

- Soybean:
  - ✓ Partly produced domestically
  - ✓ Largely **imported**
- Soybean products (Soybean oil):
  - ✓ Large share of **consumer spending**
- Result:
  - ✓ Soybean price rise → **CPI increases more than GDP deflator**

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The GDP Deflator versus the Consumer Price Index

Difference 2 – Weighting Method

- ❖ CPI:
  - Uses a **fixed basket of goods and services**
  - Basket updated **infrequently**
- ❖ GDP deflator:
  - Uses **current production basket**
  - Basket **changes automatically over time**

Implication of Weighting Difference

- o If all prices change proportionately:
  - ✓ Difference is **not important**
- o If prices change unevenly:
  - ✓ Weighting method **matters significantly**
- o Therefore:
  - ✓ Inflation rates may **differ between CPI and GDP deflator**

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Numerical Example – CPI vs GDP Deflator

Step 1: Economy Setup

- Assume an economy produces only two goods:

Good	Quantity (Base Year)	Price (Base Year)	Quantity (Current Year)	Price (Current Year)
Rice (domestic)	100	10	120	12
Car (imported)	10	100	8	150

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Numerical Example – CPI vs GDP Deflator

Step 2 – CPI Calculation (Fixed Basket)

- **CPI uses base-year quantities**
- Base-year basket:
  - Rice = 100 units
  - Car = 10 units
- **Cost of basket:**
  - Base year = (100×10) + (10×100) = 1000 + 1000 = **2000**
  - Current year = (100×12) + (10×150) = 1200 + 1500 = **2700**
- **CPI:**

$$CPI = \frac{2700}{2000} \times 100 = 135$$

- **Inflation (CPI): 35%**

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Numerical Example – CPI vs GDP Deflator

Step 3 – GDP Deflator Calculation

- **GDP deflator uses current production quantities**
- **Nominal GDP (current prices):**
  = (120×12) + (8×150)
  = 1440 + 1200 = **2640**
- **Real GDP (base prices):**
  = (120×10) + (8×100)
  = 1200 + 800 = **2000**
- **GDP Deflator:**
  $$= \frac{2640}{2000} \times 100 = 132$$
- **Inflation (GDP Deflator): 32%**

Step 4 – Why Results Differ

- CPI = 35%
- GDP Deflator = 32%
- Reason: Fixed vs Changing Basket
  - CPI:
    - ✓ Uses **old (base-year) quantities**
    - ✓ Still assumes **10 cars (even though consumption fell to 8)**
  - GDP Deflator:
    - ✓ Uses **current quantities**
    - ✓ Reflects **reduced car consumption**

CORRECTING ECONOMIC VARIABLES DUE TO INFLATION

Purpose of Measuring Price Level

- ❖ The purpose of measuring the **overall level of prices in the economy** is to:
  - ✓ Allow us to **compare dollar figures from different times**
- ❖ Once we know how price indexes are calculated:
  - ✓ We can use them to compare a **dollar figure from the past to a dollar figure in the present**

Dollar Figures from Different Times

- ❑ Question:
  - o Was Babe Ruth's salary of **$80,000 in 1931 high or low** compared to today's players?
- ❑ To answer:
  - o Need to know:
    - ✓ **Price level in 1931**
    - ✓ **Price level today**

CORRECTING ECONOMIC VARIABLES DUE TO INFLATION

Dollar Figures from Different Times

Why Adjustment is Needed

- Part of the increase in salaries:
  - ✓ Compensates for **higher prices today**
- Therefore:
  - ✓ To compare salaries across time:
    - – We must **inflate past values into today's dollars**

Inflation Adjustment Formula

❑ **General Formula:**

$$\text{Amount } \textit{in today's dollars} = \textit{Amount in year T dollars} \times \frac{\textit{Price level today}}{\textit{Price level in year T}}$$

- ✓ A price index such as the **CPI measures the price level**
- ✓ It determines the **size of the inflation correction**

CORRECTING ECONOMIC VARIABLES DUE TO INFLATION

Dollar Figures from Different Times

Example – Babe Ruth's Salary

- Salary in 1931 = **$80,000**
- CPI in 1931 = **15.2**
- CPI in 2015 = **237**
- Salary *in 2015 dollars* $= 80{,}000 \times \frac{237}{15.2}$

$$= 80{,}000 \times 15.6 = 1{,}247{,}368$$

- Equivalent salary:
  - **$1,247,368 (≈ $1.2 million)**

Interpretation

- Babe Ruth's salary: Was a **good income**
- But:
  - ✓ Only about **one-third of the average player's salary today**
  - ✓ Only about **4% of Clayton Kershaw's salary**

CORRECTING ECONOMIC VARIABLES DUE TO INFLATION

Dollar Figures from Different Times

Second Example – President Hoover

- Salary in 1931 = **$75,000**
- **Conversion:**

$$75{,}000 \times \frac{237}{15.2} = 1{,}169{,}408$$

- Equivalent salary in 2015: **$1,169,408**

Comparison Insight

- Hoover's adjusted salary:
  - ✓ Much higher than:
    - – President Obama's salary = **$400,000**
- Conclusion:
  - ✓ Hoover had a **very high real income**

CORRECTING ECONOMIC VARIABLES DUE TO INFLATION

Mr. Index Goes to Hollywood

- ❑ Movie popularity is usually measured by:
  - **Box office receipts (nominal)**
- ❑ Problem:
  - Prices (ticket prices) **rise over time**
- ❑ Inflation gives:
  - **Advantage to newer films**

Explanation

- Older movies: Had **lower ticket prices (e.g., $0.25)**
- But: Much larger audiences (e.g., **90 million weekly viewers**)
- Inflation adjustment reveals: **True popularity comparison**

Example – Movie Rankings

- ❖ **Without inflation adjustment:**
  - *Star Wars: The Force Awakens (2015)* → #1
  - *Avatar (2009), Titanic (1997)* follow
- ❖ **With inflation adjustment:**
  - *Gone with the Wind (1939)* → #1
  - Followed by:
    - Original *Star Wars* (1977)
    - *The Sound of Music* (1965)